

Java DSA + Backend 4-Month Study Plan

Pepcoding as the main DSA understanding source. Kunal Kushwaha as the missing interview, OOP, and modern Java add-on source. LeetCode/GFG as the final practice source.

Main Rule
Do not randomly jump between playlists. Follow Pepcoding in a controlled order, and wherever Pepcoding is weak or missing for fresher Java backend jobs, add selected Kunal lectures.

1. Source Strategy

- Pepcoding = main DSA understanding source.
- Kunal Kushwaha = missing interview/OOP/modern Java add-on source.
- LeetCode/GFG = final practice source.
- Kunal is not the main course now; Kunal is the gap-filler and interview booster.

2. Total Video Hours Calculated

Source	Videos / scope	Visible duration	Notes
Kunal Kushwaha	69 videos	79h 35m 37s	Java + DSA + interview preparation course; repo includes code samples, assignments, and notes.
Pepcoding raw visible total	All pasted Pepcoding material	84h 47m 49s	Raw total counts repeated pasted entries.
Pepcoding cleaned total	Duplicate Binary Tree repeats removed	81h 59m 41s	More useful planning total. Some first videos had no visible duration, so real total may be slightly higher.

3. Pepcoding Playlist-Wise Duration

Pepcoding playlist	Visible videos counted	Duration
Java Foundation	74	6h 7m 29s
Functions & Arrays	43	3h 59m 45s
2D Arrays	19	2h 16m 36s
Recursion & Backtracking Level 1	53	8h 37m 27s
Time & Space	14 visible durations	3h 21m 9s
Linked List	53 visible durations	3h 19m 4s
Stack & Queue	56	6h 58m 4s
Generic Tree	52 visible durations	5h 32m 43s
Binary Tree	raw 61 / cleaned 43	10h 43m raw / 7h 55m cleaned
BST	17 visible durations	1h 28m 40s
HashMap & Heap	22 visible durations	3h 39m 34s
Dynamic Programming	35 visible durations	11h 54m 29s
Graphs	17 visible durations	5h 53m 39s
Backtracking Level 2	38 visible durations	10h 55m 35s

Pepcoding's DSA Level 1 playlist is publicly organized around Java Foundation, Functions/Arrays, Strings/StringBuilder/ArrayList, and DSA Level 1 modules.

4. Final Correct Structure

- 1 Java Foundation
- 2 Functions + Arrays
- 3 2D Arrays

- 4 Searching + Binary Search
- 5 Sorting + Time Complexity
- 6 Strings + StringBuilder + ArrayList
- 7 Basic Maths + Bit Manipulation
- 8 Recursion Level 1
- 9 Backtracking basics
- 10 OOP + Core Java
- 11 Linked List
- 12 Stack + Queue
- 13 HashMap
- 14 Heap / PriorityQueue
- 15 Generic Tree
- 16 Binary Tree
- 17 Binary Search Tree
- 18 Dynamic Programming
- 19 Graphs
- 20 Backtracking Level 2 selected problems
- 21 Interview revision + mocks

5. Playlist-Wise Exact Plan: Keep, Skip, Add From Kunal

Phase 1: Java Foundation

Source: Pepcoding Java Foundation playlist

Use Pepcoding as the main source for basics, loops, input, number problems, GCD/LCM, prime, Fibonacci, and patterns.

Area	What to do
Keep / complete	variables; conditionals; loops; input; prime number; Fibonacci; count digits; digits of number; reverse number; inverse number; rotate number; GCD/LCM; prime factorization
Remove / reduce / optional	too many pattern questions; Print Z; repeated beginner syntax videos if already clear; do only 5-6 pattern questions, not all 20
Add from Kunal	Kunal 5: Introduction to Programming, only if weak; Kunal 6: Flowcharts & Pseudocode, only if weak; Kunal 8: First Java Program, I/O, Debugging, Datatypes, only if weak; Kunal 9: Conditionals and Loops, only if weak; Kunal 11: Functions/Methods, only if weak
Must-cover / notes	Do not spend too much time here.

Phase 2: Functions + Arrays

Source: Pepcoding Functions and Arrays Level 1

Use Pepcoding for foundation-heavy array clarity, then add Kunal for interview patterns.

Area	What to do
Keep / complete	functions; number system basics; arrays introduction; memory; span of array; find element; sum of two arrays; difference of two arrays; reverse array; rotate array; inverse array; subarrays; subsets; binary search; ceil/floor; first/last index
Remove / reduce / optional	base addition/subtraction/multiplication if number systems are already clear; bar chart, only if you want loop practice
Add from Kunal	Kunal 12: Arrays and ArrayList; Kunal 13: Linear Search; Kunal 14: Binary Search; Kunal 15: Binary Search Interview Questions; Kunal 16: Binary Search in 2D Arrays
Must-cover / notes	two pointers; sliding window basics; prefix sum; Kadane's algorithm; Dutch national flag; binary search on answer; rotated sorted array; peak element

Phase 3: 2D Arrays

Source: Pepcoding 2D Arrays Level 1

Build matrix traversal and implementation confidence without overinvesting in low-frequency problems.

Area	What to do
Keep / complete	matrix multiplication; wave traversal; spiral traversal; exit point; rotate 90 degree; shell rotate; diagonal traversal; saddle point; search in sorted 2D array
Remove / reduce / optional	do not spend too much time on shell rotate; understand once and move on
Add from Kunal	Kunal 16: Binary Search in 2D Arrays
Must-cover / notes	This add-on is enough for this phase.

Phase 4: Time & Space + Sorting

Source: Pepcoding Time and Space Level 1

Use Pepcoding for sorting algorithms and complexity intuition, then add Cycle Sort for interview patterns.

Area	What to do
Keep / complete	bubble sort; selection sort; insertion sort; merge two sorted arrays; merge sort; partitioning; quick sort; quick select; count sort; radix sort; sort 01; sort 012; target sum pair; pivot of sorted rotated array
Add from Kunal	Kunal 20: Cycle Sort; Kunal 24: Time and Space Complexity; Kunal 59: Count Sort; Kunal 60: Radix Sort
Must-cover / notes	Cycle Sort helps with missing number, duplicate number, set mismatch, and first missing positive.

Phase 5: Strings, StringBuilder, Maths, Bit Manipulation

Source: Kunal as the strong add-on source

Pepcoding Level 1 can feel incomplete here for modern interviews, so make Kunal the main source for this phase.

Area	What to do
Add from Kunal	Kunal 21: Strings and StringBuilder; Kunal 25: Bitwise Operators + Number Systems; Kunal 26: Maths for DSA; Kunal 52: StringBuffer; Kunal 53: BigInteger & BigDecimal
Must-cover / notes	String immutability; StringBuilder; palindrome; anagram; frequency count; substring logic; XOR; power of two; find unique element; GCD/LCM; sieve; modular arithmetic basics; This phase is important for Java backend and coding rounds.

Phase 6: Recursion Level 1

Source: Pepcoding Recursion and Backtracking Level 1

Pepcoding is the main source for recursion basics, recursion in arrays, subsequences, keypad, paths, permutations, encodings, flood fill, target sum, N-Queens, and Knight's Tour.

Area	What to do
Keep / complete	print increasing/decreasing; factorial; power linear/log; print zig-zag; Tower of Hanoi; recursion in arrays; first/last/all indices; get subsequence; keypad; stair paths; maze paths; print subsequence; print permutations; print encodings; flood fill; target sum subsets; N-Queens
Remove / reduce / optional	Knight's Tour is optional for now; too many maze variations; do enough to understand the pattern
Add from Kunal	Kunal 23: Intro to Recursion; Kunal 27: Recursion Level 1 Questions; Kunal 28: Recursion Array Questions; Kunal 32: Subset/Subsequence/String Questions; Kunal 33: Permutations; Kunal 35: Backtracking Maze; Kunal 36: N-Queens, N-Knights, Sudoku
Must-cover / notes	Use Kunal mainly for interview mapping, not as a full second course.

Phase 7: OOP + Core Java

Source: Kunal OOP series

This is missing as a proper deep module in Pepcoding and is compulsory for a Java backend shift.

Area	What to do
Add from Kunal	Kunal 37: OOP 1; Kunal 38: OOP 2; Kunal 39: OOP 3; Kunal 40: OOP 4; Kunal 41: OOP 5; Kunal 42: OOP 6; Kunal 43: OOP 7
Must-cover / notes	classes; objects; constructors; static; packages; inheritance; polymorphism; encapsulation; abstraction; interfaces; generics; lambda; exception handling; collections; Vector; enums; This phase is mandatory because the target is Java backend development, not only DSA problem solving.

Phase 8: Linked List

Source: Pepcoding Linked List Level 1

Use Pepcoding for implementation and core operations, then add Kunal for interview bridging.

Area	What to do
Keep / complete	implementation; add/remove/get; reverse data iterative; reverse pointer iterative; kth from end; middle; merge two sorted lists; merge sort linked list; remove duplicates; odd-even; k reverse; recursive reverse; palindrome; fold; add two lists; intersection point
Add from Kunal	Kunal 44: Linked List Tutorial; Kunal 45: Linked List Interview Questions
Must-cover / notes	cycle detection; cycle starting node; remove nth node; reorder list; add two numbers; reverse in k-group, optional

Phase 9: Stack and Queue

Source: Pepcoding Stack and Queue Level 1

Focus on bracket, monotonic stack, expression, interval, and adapter problems.

Area	What to do
Keep / complete	duplicate brackets; balanced brackets; next greater element; stock span; largest rectangle histogram; sliding window maximum; infix evaluation; infix conversion; celebrity problem; postfix/prefix evaluation; merge overlapping intervals; min stack; queue/stack adapters
Remove / reduce / optional	too many adapter variations if implementation is already clear; OOP mini swap examples because Kunal OOP is already included
Add from Kunal	Kunal 46: Stacks and Queues Tutorial; Kunal 47: Stacks and Queues Interview Questions

Phase 10: HashMap and Heap

Source: Pepcoding HashMap and Heaps Level 1

Do this before trees because it is highly useful in interviews.

Area	What to do
Keep / complete	hashmap intro; highest frequency character; common elements; longest consecutive sequence; heap intro; k largest elements; nearly sorted array; median priority queue; merge k sorted lists; implement priority queue; implement hashmap; comparable vs comparator
Add from Kunal	Kunal 56: Heap + PriorityQueue; Kunal 57: HashMap & HashTable
Must-cover / notes	two sum; group anagrams; top k frequent elements; subarray sum equals k; longest consecutive sequence; kth largest element; merge k sorted lists

Phase 11: Generic Tree

Source: Pepcoding Generic Trees Level 1

No major Kunal add-on is needed here.

Area	What to do
Keep / complete	construction; display; size/max/height; traversals; level order; mirror; remove leaves; linearize; find; node to root path; LCA; distance between nodes; similar/mirror/symmetric; multisolver; predecessor/successor; ceil/floor; kth largest; diameter; iterative preorder/postorder
Remove / reduce / optional	Iterable and Iterator optional for now; advanced generic-tree problems if time is low

Phase 12: Binary Tree

Source: Pepcoding Binary Trees Level 1

The pasted Binary Tree list includes repeated entries. Use the cleaned useful content, roughly 7h 55m.

Area	What to do
Keep / complete	constructor; display; size/sum/max/height; traversals; level order; iterative traversals; node to root path; print K levels down; print nodes K distance away; path to leaf; remove leaves; diameter; tilt; is BST; balanced tree; largest BST subtree; construct tree from preorder/inorder; construct from postorder/inorder; bottom view
Remove / reduce / optional	duplicate repeated entries in the pasted list
Add from Kunal	Kunal 49: Binary Trees + BST tutorial; Kunal 55: Binary Tree Interview Questions; Kunal 63: Binary Tree from Preorder & Inorder; Kunal 64: Vertical Order Traversal; Kunal 66: Two Sum IV; Kunal 67: Kth Smallest in BST; Kunal 68: Convert Binary Tree to DLL; Kunal 69: Correct Swapped Nodes in BST

Phase 13: BST

Source: Pepcoding Binary Search Tree Level 1

Use Pepcoding for BST fundamentals, then add targeted Kunal interview problems.

Area	What to do
Keep / complete	constructor; size/sum/max/min; add node; remove node; replace with sum of larger; LCA; print in range; target sum pair
Add from Kunal	Kunal 67: Kth Smallest in BST; Kunal 69: Correct Swapped Nodes in BST

Phase 14: Dynamic Programming

Source: Pepcoding Dynamic Programming Level 1

Kunal's 69-video list does not show a full dedicated DP module, so keep Pepcoding as the main source.

Area	What to do
Keep / complete	introduction to DP; climbing stairs; climbing stairs with jumps; min moves; min cost path; goldmine; target sum subset; coin change combination; coin change permutation; 0/1 knapsack; unbounded knapsack; count binary strings; arrange buildings; decode ways; max sum non-adjacent; paint house; tiling; friends pairing; partition into subsets; stock buy/sell variants
Remove / reduce / optional	Highway Billboard optional; very hard stock K transactions after basics

Phase 15: Graphs

Source: Pepcoding Graphs Level 1

Pepcoding is stronger as the main source for graphs.

Area	What to do
Keep / complete	graph representation; has path; all paths; DFS; connected components; is connected; number of islands; BFS; cycle detection; bipartite; infection spread; Dijkstra; Prim's; topological sort; iterative DFS
Remove / reduce / optional	Hamiltonian path/cycle optional for first pass; Knight's Tour already optional from recursion
Add from Kunal	Kunal 65: Word Ladder

Phase 16: Backtracking Level 2

Source: Pepcoding Backtracking Level 2

This is not fully compulsory for the first job-prep pass. Do selected problems before applications.

Area	What to do
Keep / complete	abbreviations; N-Queens branch and bound; Sudoku; palindrome partitioning; word break; remove invalid parentheses; permutations; combinations; coin change recursion; equal sum subset; maximum number after K swaps, optional
Remove / reduce / optional	crossword puzzle; cryptarithmic; too many queen combination/permutation variations; too many words K-selection variations; Tug of War optional; advanced partition K subsets optional
Must-cover / notes	Backtracking Level 2 is almost 11 hours, so do not complete everything before job applications.

6. Kunal Lectures You Should Definitely Add

Kunal lecture	Why add
12 Arrays and ArrayList	Java ArrayList clarity
14 Binary Search	Better interview explanation
15 Binary Search Interview Questions	Very important
20 Cycle Sort	Missing interview pattern
21 Strings and StringBuilder	Important for Java
24 Time and Space Complexity	Strong Big-O clarity
25 Bitwise + Number System	Pepcoding not enough for interview
26 Maths for DSA	Useful for coding rounds
37-43 OOP series	Mandatory for Java backend
45 Linked List Interview Questions	Interview bridge
47 Stack/Queue Interview Questions	Interview bridge
55 Binary Tree Interview Questions	Very important
56 Heap/PriorityQueue	Java PriorityQueue clarity
57 HashMap/HashTable	Java HashMap clarity
63-69 Tree/BST interview problems	Good revision

7. Kunal Lectures You Can Skip for Now

Kunal lecture	Reason
1-3 intro/interview motivation	Watch only if you want motivation
4 Git/GitHub	Important, but not DSA; do separately
22 Pattern Questions	Pepcoding already has patterns
48 Tic Tac Toe	Not needed for DSA/job prep
50 AVL Tree	Optional
51 Segment Tree	Optional for now
54 File Handling	Java useful, but not DSA priority
58 Karp-Rabin	Optional
61 Huffman Coding	Skip for fresher DSA
62 Mo's Algorithm	Skip for now

8. Final 4-Month Execution Plan

Month	Complete	Kunal add-ons	Target
Month 1	Java Foundation fast revision; Functions & Arrays; 2D Arrays; Searching; Sorting; Strings; Time complexity; Bit manipulation basics.	Binary Search; Cycle Sort; Strings; Time Complexity; Bitwise.	90-100 questions

Month	Complete	Kunal add-ons	Target
Month 2	Recursion Level 1; Backtracking basics; OOP from Kunal; Linked List.	OOP 37-43; Recursion subset/permutation if needed; Linked List interview questions.	180-200 total questions
Month 3	Stack/Queue; HashMap; Heap; Generic Tree; Binary Tree.	Stack/Queue interview questions; Heap/PriorityQueue; HashMap; Binary Tree interview questions.	280-300 total questions
Month 4	BST; DP Level 1 selected; Graphs Level 1 selected; Backtracking Level 2 selected; mocks and revision.	BST/tree interview problems and selected graph problem Word Ladder.	350-400 total questions + job applications

9. Final Follow Format

1. Pepcoding Java Foundation
2. Pepcoding Functions & Arrays + Kunal Binary Search + Cycle Sort
3. Pepcoding 2D Arrays + Kunal Binary Search in 2D
4. Pepcoding Time & Space + Kunal Time Complexity
5. Kunal Strings + Bitwise + Maths
6. Pepcoding Recursion Level 1 + Kunal Recursion selected
7. Kunal OOP 1-7
8. Pepcoding Linked List + Kunal Linked List Interview Questions
9. Pepcoding Stack & Queue + Kunal Stack/Queue Interview Questions
10. Pepcoding HashMap & Heap + Kunal HashMap + Heap
11. Pepcoding Generic Tree
12. Pepcoding Binary Tree + Kunal Binary Tree Interview Questions
13. Pepcoding BST + Kunal BST interview questions
14. Pepcoding DP Level 1 selected
15. Pepcoding Graphs Level 1 selected
16. Pepcoding Backtracking Level 2 selected
17. LeetCode/GFG revision + mocks

Final Rule

Pepcoding remains the main learning source because you understand it better. Kunal is the gap-filler and interview booster. Finish with LeetCode/GFG revision and mocks.